

Project Design Phase-II
Data Flow Diagram & User Stories

Date	9 th march 2026
Team ID	EL2026TMID10104
Project Name	AI-Based Anomaly Detection Platform
Maximum Marks	4 Marks

Data Flow Diagrams:

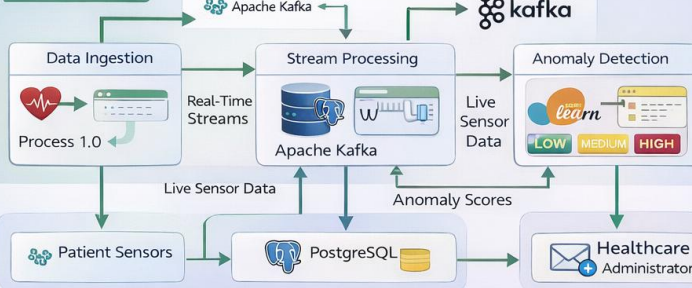
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

AI-Driven Healthcare Anomaly Detection System Data Flow Diagrams

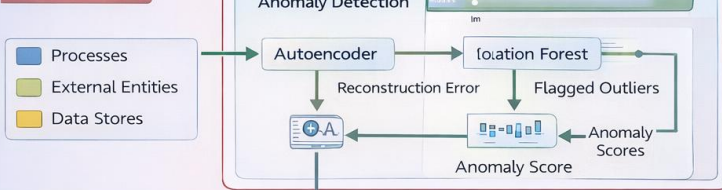
DFD Level 0



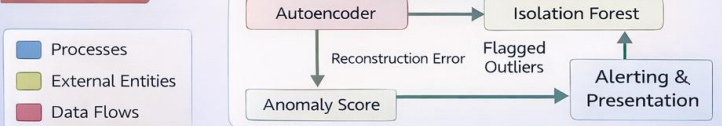
DFD Level 1



DFD Level 2



DFD Level 2



User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Customer (Mobile user)	Registration	USN-1	As a user, I can register for the system by entering email, password and confirming password.	User account is created and user can access dashboard.	High	Sprint-1
Customer (Mobile user)	Registration	USN-2	As a user, I receive a confirmation email after successful registration.	User receives email and verifies account.	High	Sprint-1
Customer (Mobile user)	Login	USN-3	As a user, I can login using email and password.	User successfully logs into system.	High	Sprint-1
Customer (Mobile user)	Dashboard	USN-4	As a user, I can view my real-time health vitals such as heart rate, SpO ₂ , temperature and blood pressure on the dashboard.	Dashboard shows live vitals data.	High	Sprint-1
Customer (Mobile user)	Health Monitoring	USN-5	As a user, I can monitor my health status categorized as Normal, Low, Medium or High risk.	System displays health status based on anomaly score.	High	Sprint-2
Customer (Mobile user)	Alerts	USN-6	As a user, I receive alerts when abnormal health conditions are detected.	User receives alert notification when anomaly occurs.	High	Sprint-2
Customer (Web user)	Dashboard Visualization	USN-7	As a healthcare staff member, I can view real-time patient data through a web dashboard.	Dashboard displays patient vitals and anomaly status.	High	Sprint-2
Customer (Web user)	Historical Data	USN-8	As a healthcare staff member, I can view historical patient health data.	System displays stored patient records from database.	Medium	Sprint-2
Customer Care Executive	Alert Monitoring	USN-9	As a healthcare executive, I can monitor alerts generated by the anomaly detection system.	Alerts are visible in monitoring panel.	High	Sprint-3

Administrator	Anomaly Detection	USN-10	As an administrator, I want the system to analyze patient vitals using machine learning models.	Autoencoder and Isolation Forest detect anomalies.	High	Sprint-2
Administrator	Email Alert System	USN-11	As an administrator, I want critical anomalies to trigger automatic email alerts.	Email notification sent for HIGH severity alerts.	High	Sprint-2
Administrator	Data Storage	USN-12	As an administrator, I want patient data and anomaly scores stored in database.	PostgreSQL database stores patient vitals and alerts.	Medium	Sprint-3